Reproducibility and Numerical Conditioning of the Drain-Refined Impact-Ionization Trio

Liang–1997 SHE–BTE 0.35 μm LDD n-MOSFET reproduction

Cross-platform convergence study — observable-based certification (interim)

Technical report for independent review

2026-10-21 (frozen solver tag `solver-frozen-2026-09-26`, git `930dd86`)

Executive summary

The certified reproduction of Liang *et al.* (1997) yields a drain-refined spatial-convergence trio of the peak impact-ionization generation rate G_{ii} at three meshes (71×58, 100×70, 141×85 on a common frozen energy grid). This report documents the reproducibility campaign and its reviewer-approved corrected-methodology confirmation, and it incorporates an independent referee's "major revisions" feedback. Four findings:

1. **The qualitative verdict is robust and reproduces everywhere tested**: the trio is significantly *non-monotone* ($S_{\mathrm{rev}} \gg 1$, a heuristic sign-reversal diagnostic, not a statistical score), so a conventional one-term Richardson extrapolation is not justified from these three points, and **there is no established continuum value of G_{ii}**.
2. **All three meshes certify to $<1\%$ under the same solver.** 71×58 +0.006% and 100×70 +0.40% are from the current corrected-methodology run (git `4e2374e`); 141×85 +0.79% is from an *earlier* run (git `1893b9c`) whose only source difference is a monitoring script (zero physics/numerics), hence solver-equivalent. The large mid-investigation $\sim$10–12% platform differences were finite-iteration (premature-stop) artifacts, not converged discrepancies.
3. **The reproducible quantity is the physical observable, not the outer residual.** A diagnostic continuation (each converged state continued 100 iterations with stopping *disabled*) shows the observables (G_{ii}, $\int G_{ii}\, dV$, drain current) are flat to $<0.35\%$, while the outer residual R_ϕ is itself ill-conditioned and *does not stay* below the 10^{-4} tolerance (it oscillates in a mesh-dependent band mostly *above* tolerance). A single R_ϕ crossing is therefore not a sufficient certification; observable flatness is. This confirms the referee's central methodological point and strengthens the reproducibility conclusion.
4. **Convergence difficulty sharpens with mesh refinement.** Iterations-to-R_ϕ rose 325 $\rightarrow$ 929 ($71\times58 \rightarrow 100\times70$); the R_ϕ noise floor rose correspondingly ($\sim 1.8\times10^{-4} \rightarrow \sim 4.5\times10^{-4}$); and on the finest mesh 141×85 the current run's *inner* (Krylov/ILU) linear solve stagnated and the outer iteration hung (classified *unresolved* for that run). The finest mesh is the fragile one: one run hung, an earlier run certified at +0.79%.

Peak G_{ii} is a near-null-space, high-energy-tail functional, hypersensitive to the platform floating-point path. A staged frozen-state probe (Appendix C) localizes the *first* cross-platform divergence to sparse-system assembly ($\sim 10^{-16}$, single-threaded), upstream of the solve. This rounding-scale seed perturbs the convergence *path, rate, and residual*, but not the converged observable. The recommended methodology is to certify on sustained observable flatness (and report a well-conditioned

integrated observable and a properly-defined terminal current), rather than on a single outer-residual crossing or the over-strict pointwise plateau gate.

1 System, reference, and method

First-order spherical-harmonic expansion (SHE) of the electron Boltzmann equation in total-energy form, self-consistent with 2D Poisson and hole continuity; nonparabolic Kane band; DOS-weighted acoustic + optical-phonon + Keldysh-surrogate impact-ionization operators. The SHE linear system is solved with LGMRES preconditioned by a physics-split incomplete-LU (`scipy.sparse.linalg.spilu`); the outer map is Anderson-accelerated (depth = 6, $\beta = 0.5$). The outer fixed-point is stopped on the potential residual $R_\phi = \max|G(\phi) - \phi|$ over free DOFs (the raw fixed-point residual, not the damped Anderson step), tolerance $\mathrm{tol}_\phi = 10^{-4}$. A legacy acceptance gate (last-4 cumulative-iteration spread of $G_{ii,\max}$ and $G_{tot} \le 3 \times 10^{-4}$ with $R_\phi \le 1.5 \times 10^{-3}$) is retained as a diagnostic only; §3 shows both R_ϕ and that gate are inadequate as sole certification criteria.

Certified reference (Sep-24, Windows, pure-Python kernel, git `d30b471`, equivalent to the frozen tag):

$$G_{ii} = 3.075619/2.136180/2.387429 \times 10^{25}\ \mathrm{cm^{-3}s^{-1}}, \quad S_{\mathrm{rev}} = 737,\ \text{non-monotone.}$$

Common frozen H-grid hash `b4319bb03303` (591 energy points) across all meshes and runs. This reference was *plateau*-certified under the legacy gate; its per-node outer residual is not uniformly $< 10^{-4}$ (see §3), which is addressed by the observable-based criterion below.

2 Cross-platform R_ϕ-convergence confirmation (the approved run)

Each mesh was run continuously to $\mathrm{tol}_\phi = 10^{-4}$ on Linux (matched stack: NumPy 2.5.1 / SciPy 1.18.0 / Python 3.13.12, pure-Python), with the 1500-iteration cap as a safety limit only, nodes warm-chained.

mesh	n_{iter}	R_ϕ at stop	G_{ii} ($\times 10^{25}$)	vs Windows ref	provenance
71×58	325	7.10×10^{-5}	3.075752	+0.006%	current run (`4e2374e`)
100×70	929	6.97×10^{-5}	2.144818	+0.40%	current run (`4e2374e`)
141×85	717	7.30×10^{-5}	2.406344	+0.79%	earlier run (`1893b9c`)[†]

[†]Verified via the npz `git_commit` field. The 141×85 certified state is from an earlier run instance; the *only* source difference between `1893b9c` and `4e2374e` is `src/trend_snapshot.py` (a monitoring sampler, zero physics/numerics), so it is solver-equivalent and the value is valid. It is *not*, however, the current run's node 3 (see below), so this is **not** a clean single-run three-mesh trio.

The 100×70 node is the decisive in-run demonstration of the premature-stop mechanism. Its history: a flat plateau near 1.90×10^{25} (with G_{ii} drifting slightly *downward*, $1.96 \to 1.90$) and R_ϕ oscillating in the 5×10^{-4}–2.4×10^{-3} band from iter ~ 50 to ~ 650; then a climb over iters ~ 700–894 to the reference, certifying at 929 ($R_\phi\ 7.0 \times 10^{-5}$). A fixed-iteration harness stopping at ~ 360 (as the original trio harness did) would have recorded $\sim 1.90 \times 10^{25}$ (-11%) and mis-read the down-drift as divergence — precisely the artifact that produced the withdrawn -12%.

The current run's own 141×85 did not certify. Warm-chained from the current certified 100×70, it advanced to outer iteration 281 and then **hung**: the process remained compute-bound (6.6 cores, 47 threads) for ~ 11.5 h on a single outer step while the log stayed frozen — i.e. the *inner* LGMRES/ILU linear solve stagnated (cycling without returning). It was killed and is classified

unresolved for the current run. This is a genuine finite-mesh robustness failure, not a plateau: on 100×70 the iteration kept advancing; on 141×85 it stopped advancing entirely. The earlier-run certification (+0.79%) shows the finest mesh *can* converge under the same solver, but it is slow and fragile — the near-null-space conditioning taken to its limit.

3 Diagnostic continuation: R_ϕ is noisy, the observable is converged

To test whether a single $R_\phi < 10^{-4}$ crossing certifies convergence (the referee's principal concern), each converged state was continued 100 iterations with stopping *disabled* ($\text{tol}_\phi = 0$), single-threaded, and the outer residual was independently recomputed at the saved state. Windows (reference states):

mesh	OS	R_ϕ recompute	R_ϕ med (iters $> 10^{-4}$)	$\langle G_{ii} \rangle$ ($\times 10^{25}$)	G_{ii} drift	I_d (A/μm)
71×58	Win	3.94×10^{-5}	1.8×10^{-4} (90/100)	3.07934	0.185%	3.0032×10^{-4}
71×58	Linux	7.10×10^{-5}	1.6×10^{-4} (82/100)	3.07905	0.181%	3.0032×10^{-4}
100×70	Win	2.57×10^{-4}	4.5×10^{-4} (100/100)	2.13790	0.31%	2.955×10^{-4}
100×70	Linux	6.97×10^{-5}	3.8×10^{-4} (99/100)	2.14324	0.21%	2.951×10^{-4}

Cross-platform, the *observable* agrees: $\langle G_{ii} \rangle$ to 0.0096% (71×58) and 0.25% (100×70), and drain current to $< 0.15\%$ on both meshes. The *residual* does not: on 100×70 the Windows reference parks at $R_\phi = 2.57 \times 10^{-4}$ while Linux stopped at 6.97×10^{-5} — a $\sim 4\times$ difference against a 0.25% observable. Both platforms show R_ϕ oscillating mostly *above* 10^{-4} (82–100 of 100 iters) with the observables flat to $< 0.35\%$. The two sides are symmetric and corroborate each other.

The Linux-side diagnostic (matched stack) was run identically and is included above; it reproduces the Windows shape on both meshes.

Reading.

1. **A single R_ϕ crossing is not sustained convergence (referee confirmed), on both platforms.** On 71×58, 90/100 (Windows) and 82/100 (Linux) continued iterations sit *above* 10^{-4}; the stop had caught a transient dip. On 100×70, the Windows reference state recomputes to $R_\phi = 2.57 \times 10^{-4}$ — it never satisfies 10^{-4} at all (100/100 above) — consistent with its plateau-certification rather than an $R_\phi < 10^{-4}$ certification.
2. **But the observables are converged.** G_{ii}, G_{tot}, and drain current are flat to $< 0.35\%$ across all 100 iterations on both meshes. The state is parked at the converged solution; only the residual jitters.
3. **Decisive cross-platform datum.** Linux 100×70 stopped at $R_\phi = 6.97 \times 10^{-5}$ (a dip), the Windows reference parks at $R_\phi = 2.57 \times 10^{-4}$: the residuals differ by $\sim 4\times$, yet the observables agree to 0.40%. The observable is the reproducible converged quantity; R_ϕ is an ill-conditioned metric.
4. **The residual noise floor rises with refinement** (median $1.8 \times 10^{-4} \to 4.5 \times 10^{-4}$ from 71×58 to 100×70, and inner-solve stagnation on 141×85) — residual-level evidence for the conditioning sharpening with mesh density.

Consequence for certification: certify on sustained observable flatness (e.g. G_{ii}/G_{tot} drift over a fixed window), not on a single R_ϕ crossing nor the over-strict pointwise plateau gate.

4 The prior campaign (finite-iteration snapshots)

The columns below (except as noted) are *finite-iteration snapshots* at a fixed stopping point; they document the campaign that produced and then overturned the apparent platform discrepancy.

node	Sep-24 ref	Win intr.	Win clean	Lx numba	Lx 1.18[‡]	**Lx R_ϕ-conv.**
71×58	3.075619	3.075619	3.075619	2.9847	2.7067	**3.075752**
100×70	2.136180	2.142249	2.087561	∼1.80[§]	—	**2.144818**
141×85	2.387429	2.375921	2.399434	—	—	**2.406344**[†]

[‡]matched-stack snapshot at ∼360 iters, did not certify. [§]hotspot limit-cycle at a fixed budget, never certified — *resolved to* 2.144818 *at R_ϕ convergence.* [†]earlier-run (solver-equivalent) node, §2. Every figure except the last column is a snapshot, not a converged value.

The non-reproducibility "hierarchy" was a convergence-rate hierarchy. The mid-investigation deltas (gate-trip ∼ 2%, kernel ∼ 3–7%, SciPy ∼ 3%, OS ∼ 12%) were all finite-iteration snapshots. The matched-stack 71×58 read 2.7067×10^{25} (-12%) at 360 iters; continued to convergence the same configuration reaches 3.0758×10^{25} ($+0.006\%$). Matching the SciPy version moved the 360-iteration snapshot the *wrong* way ($-9.1\% \to -12.0\%$), correctly ruling out the SciPy version as a *sufficient* cause — it changed the transient, not the endpoint.

5 Mechanism

Peak G_{ii} is numerically ill-conditioned: (i) a tail-dominated functional ($\varepsilon \gtrsim 1.5\,\mathrm{eV}$) controlled by a tiny near-null-space of the solution (a 10^{-8} linear residual does not pin it); (ii) floating-point-path-dependent assembly and solves (Appendix C: assembly coefficients differ across OS at $\sim 10^{-16}$, single-threaded, from vectorized `exp`/`pow` in the coefficient kernels); (iii) amplification over hundreds of outer iterations. These compound along the ill-conditioned direction, producing large *finite-iteration* deviations, a noisy outer residual (§3), slow and variable convergence (Finding 4), and on the finest mesh an inner-solve stagnation — but *not* a non-reproducible converged observable.

6 Remediation (proposed; none applied to the frozen reference)

- **Certify on the observable, not the residual.** Require sustained flatness of G_{ii} (and $\int G_{ii}\,dV$) over a fixed window; report R_ϕ alongside as a diagnostic with its (mesh-dependent) noise floor. A single R_ϕ crossing is insufficient (§3).
- **Run each mesh to its own convergence**, not a common iteration budget; the finest meshes need a larger budget (and may still stagnate — see direct solve).
- **Direct linear solve** where memory permits (raise the direct-solve DOF cutoff), removing the Krylov/ILU floating-point-path dependence and the inner-solve stagnation seen on 141×85. *This is a method change and belongs in a separate error-budget study, not the frozen-method trio.*
- **Report a well-conditioned integrated observable and a properly-defined terminal current.** Quote $\int G_{ii}\,dV$ (generation rate per out-of-plane width) and the substrate hole current at the body contact including recombination (audit F-08); note $\int G_{ii}\,dV$ is *not* itself the collected terminal current. The diagnostic already records the drain current ($\sim 3.0 \times 10^{-4}$ A/μm, drift $< 0.1\%$) and relative number balance ($\sim 10^{-4}$); the spatial convergence of $\int G_{ii}\,dV$ is *not* yet demonstrated (node-to-node G_{tot} changes $\sim 35\%$, a discretization effect distinct from floating-point conditioning).

7 For the independent reviewer (response to the "major revisions" review)

- **Accepted and incorporated:** (1) a single R_ϕ crossing is not certified convergence — now demonstrated directly (§3) and the certification basis changed to observable flatness; (2) the title/claims no longer assert a clean single-run three-mesh confirmation — node 3 is labeled an earlier-run solver-equivalent result and the current-run 141×85 is reported as unresolved (hung); (B) the Windows reference used the legacy gate, not $R_\phi < 10^{-4}$ — confirmed (100×70 reference $R_\phi = 2.57 \times 10^{-4}$), which is why the comparison is now stated on the observable; (C) a 2-mesh iteration-count increase is not a monotonic three-mesh trend — restated as supported by the residual-floor evidence; (E) spatial convergence of $\int G_{ii}\, dV$ is not demonstrated, and $G_{tot} \neq$ terminal current — both stated; (F) provenance (dirty tree, `F3d` stored `float32`) — documented below.
- **Robust (verify on Windows):** the non-monotone trio.
- **Reproducible when converged (verified both platforms):** the *observable* agrees across Windows/Linux to 0.0096% (71×58) and 0.25% (100×70) via the symmetric diagnostic (§3), and across 10^{-16} perturbations to $< 1\%$ (Appendix C). A minimal check: continue a converged 100×70 state on each OS with stopping disabled and confirm G_{ii} flat to $< 0.5\%$ while R_ϕ remains in its $\sim 10^{-4}$–10^{-3} band.
- **Open:** the finest mesh's inner-solve stagnation; spatial-convergence and physical-validation studies (separate from this numerical-reproducibility report).

8 Status and scope

The earlier $\sim 12\%$ platform conclusion is **withdrawn**. **Supported:** the non-monotone trio; slow, path-dependent convergence with difficulty rising on finer meshes (iteration count, residual floor, and finally inner-solve stagnation); cross-platform agreement of the *observable* to $< 0.5\%$ on the two meshes tested this way; all three meshes certifying to $< 1\%$ under the same solver. **Not established:** a clean single-run three-mesh trio (node 3 is an earlier solver-equivalent run; the current 141×85 is unresolved); a platform-dependent converged solution; pointwise G_{ii}-plateau or single-R_ϕ-crossing convergence; spatial (continuum) convergence of G_{ii} or $\int G_{ii}\, dV$; quantitative equivalence to the original 1997 device physics (the solver is a reconstructed/surrogate model).

Three distinct questions should not be conflated: *numerical reproducibility* (same equations $\rightarrow$ similar observables across platforms under matched stopping — substantially supported on two meshes), *discretization convergence* (grids fine enough for a continuum claim — not established; anisotropic x-only/y-only refinement needed), and *physical validity* (correct device physics and terminal currents vs Liang 1997 or experiment — a separate study).

Report two quantities. Quote G_{ii}^{obs} (at observable convergence) with the window variability $\Delta G_{ii}^{\text{tail}} = (\max - \min)/\langle\cdot\rangle$ and the R_ϕ noise band, so the outer residual is not misread as proof of pointwise convergence.

Appendix: provenance

Solver tag `solver-frozen-2026-09-26` (`930dd86`); reference `d30b471`. Confirmation nodes: 71×58/100×70 npz `git_commit=4e2374e`; 141×85 npz `git_commit=1893b9c` (dirty), warm-chained from an earlier 100×70; the sole source difference `4e2374e`↔`1893b9c` is `src/trend_snapshot.py`. The driver stores the distribution array `F3d` as `float32` and keeps full iteration trajectories in text

logs; full-precision `float64` replay states are not retained for every node (a provenance limitation for bit-exact reconstruction, flagged by the reviewer). Diagnostic harness: `audit/diag_continuation.py` (history keys `max_dphi_V`, `Gii_max_cm3s`, `G_tot_cm1s`, `relative_number_balance`, `drain_current_A_per_um`). Windows: Python 3.13.15. Linux: Python 3.13.12 (uv venv); both NumPy 2.5.1 / SciPy 1.18.0, pure-Python. H-hash `b4319bb03303`, 591 points. No thermal throttling (package 70–89 °C under the 95 °C flag); the 141×85 hang was a compute-bound inner-solve stagnation, not a thermal event.

Appendix B: Direct platform-FP probe (one-shot)

The identical 71×58 cold solve, run 3 capped outer iterations on both OSs with bit-identical mesh/energy-grid input and the *same* OpenBLAS 0.3.33: ϕ diverges at 2.7×10^{-11} (well-conditioned) while G_{ii} diverges at 6.6×10^{-4} pointwise (near-null-space), growing $5.7 \times 10^{-13} \to 2.2 \times 10^{-12} \to 1.5 \times 10^{-10}$ over 3 iters. Establishes sensitivity/amplification; Appendix C localizes the first-divergence stage.

Appendix C: Staged localization and the reversing ensemble

A frozen-state probe (`audit/fixed_state_probe.py`) froze one 71×58 state, copied identical bytes to both OSs, and assembled/solved single-threaded: state, CSR structure, H, active map all bit-identical; A.`data` and b differ at $\sim 1.1 \times 10^{-16}$ (last ULP); a single solve reproduces G_{ii}^{peak} across OS to 3.8×10^{-13} (so SuperLU/LGMRES is not the primary seed). First divergence is thus in assembly-coefficient floating point (platform NumPy/libm elementary functions), upstream of the solve. A perturbation ensemble (control + three 10^{-16} seeds) initially spanned $\sim 10\%$ at 400 iters (seed 3 at -11%), then *merged* when continued to the R_ϕ tolerance at 422–613 iters: all four within 0.66% of the Windows reference (seed 3 climbed to $+0.03\%$). This established the -12% as a premature-stop artifact; the 100×70 production node (§2) independently reproduced the late break on a second mesh. A driver-equivalence experiment (Windows, continuous vs Anderson-reset-every-60) differs by only 0.44%, rejecting Anderson reset as the dominant driver effect; the driver sensitivity is a symptom of under-convergence on the ill-conditioned problem.